Table 1: Single-run antithetic top-three set recovery ↑ for the combined 120-instance audit. Entries are means and marginal descriptive 95% paired instance-cluster intervals, with runs and fixed seeds averaged within instance. **German rows measure agreement with an approximate reference, not true accuracy.** Source: `analysis120/paper_antithetic_9cells.tex` and matching CSV.

Game	P=1	P=4	P=16	P=64
PIMA/LR	0.853 [0.818, 0.887]	0.926 [0.902, 0.950]	0.965 [0.947, 0.981]	0.981 [0.967, 0.992]
PIMA/RF	0.840 [0.816, 0.863]	0.926 [0.910, 0.940]	0.959 [0.947, 0.970]	0.975 [0.965, 0.983]
PIMA/MLP	0.638 [0.611, 0.665]	0.798 [0.773, 0.824]	0.890 [0.870, 0.909]	0.940 [0.924, 0.955]
ACS/LR	0.705 [0.658, 0.752]	0.853 [0.815, 0.887]	0.926 [0.896, 0.950]	0.962 [0.940, 0.980]
ACS/RF	0.891 [0.872, 0.909]	0.949 [0.935, 0.962]	0.977 [0.969, 0.985]	0.988 [0.981, 0.993]
ACS/MLP	0.667 [0.637, 0.698]	0.816 [0.794, 0.839]	0.902 [0.883, 0.920]	0.949 [0.935, 0.962]
GERMAN (approx.)/LR	0.838 [0.802, 0.874]	0.909 [0.880, 0.936]	0.960 [0.941, 0.977]	0.980 [0.964, 0.993]
GERMAN (approx.)/RF	0.734 [0.707, 0.761]	0.857 [0.835, 0.877]	0.926 [0.910, 0.942]	0.966 [0.954, 0.976]
GERMAN (approx.)/MLP	0.613 [0.587, 0.637]	0.793 [0.769, 0.816]	0.893 [0.875, 0.911]	0.949 [0.936, 0.962]

Table 2: Exact-game macro top-three recovery ↑ over six Pima/ACS model cells. Marginal descriptive 95% paired cluster intervals. Kernel's $P = 64$ entry is a *full-cube exact control*, not a sampled-estimator result; its collapsed interval is not a population certificate. Source: `analysis120/paper_exact_macro_controls.tex`.

Method	P=1	P=4	P=16	P=64
antithetic	0.766 [0.752, 0.779]	0.878 [0.867, 0.888]	0.937 [0.929, 0.944]	0.966 [0.960, 0.971]
iid_oneway	0.575 [0.560, 0.590]	0.757 [0.743, 0.770]	0.864 [0.854, 0.875]	0.929 [0.921, 0.937]
kernel	0.445 [0.429, 0.461]	0.710 [0.695, 0.725]	0.842 [0.829, 0.853]	1.000 [1.000, 1.000]

The expanded result is not solely a restatement of the inspected first 30. The held-out next 90 have exact-game macro recovery 0.7705 [0.7539, 0.7863] and 0.9666 [0.9598, 0.9735] at $P = 1$ and 64. The exploratory first 30 gave 0.7511 and 0.9628. These cohort summaries characterize persistence under the fixed ordering, not two independent replications or a retrospective preregistration (Appendix C).

The macro hides consequential exceptions. At $P = 64$, Pima/MLP recovers the exact set with mean 0.9400 [0.9243, 0.9547], and ACS/MLP with mean 0.9490 [0.9354, 0.9618]. Both point estimates are below 0.95, while both intervals cross it; they are inconclusive relative to that threshold, not certified failures or successes. ACS/LR also has a marginal lower endpoint below 0.95. At the other extreme, ACS/RF reaches 0.9876 [0.9808, 0.9935]. This heterogeneity makes a single mean unsuitable as a budget prescription for every fitted prediction.

4.3 Controls delimit the estimator comparison

At $P = 1$, antithetic pairing has higher observed macro recovery than the equal-row independent one-way control: 0.7656 versus 0.5747 (Table 2). The difference is 19.100 percentage points, marginal interval [18.032, 20.144], simultaneous band [14.058, 24.141]. This identifies the benefit of the pairing in these games, not a new algorithm. At $P = 64$, the corresponding 3.641-point difference has a simultaneous band spanning zero, [–1.401, 8.683]; marginal intervals alone would overstate the corrected comparison.

Plain sampled Kernel regression has lower recovery at the smaller allowances in this implementation, but its largest-budget exact control reaches 1.0000 on both datasets. It is also cheaper than the nominal permutation run at that allowance: 2,048 versus 7,184 logical model rows for Pima, and 8,192 versus 9,232 for ACS. Exact enumeration is thus an important practical option in these small games, not merely an expensive abstract target. These controls do not justify comparing our custom code favorably to optimized Kernel SHAP or Leverage SHAP. The model-class and game restrictions are as important as the apparent sampling curves.

background, player grouping, target, and downstream summary before deciding which form of agreement matters. Where a small coalition cube is affordable, the exact-control results argue for considering enumeration rather than assuming random approximation is necessary.

Statistical limits. The bootstrap quantifies instance-cluster variation conditional on fixed models and backgrounds. It is not an exact confidence procedure, can be degenerate at perfect agreement, and does not make three fixed seeds into an estimate of training-distribution uncertainty. The simultaneous bands address only the reported 63-contrast family. The precision expansion followed inspected exploratory results; retained cohorts and frozen prefixes make that history auditable, but do not erase it. For ACS, household-disjoint splits prevent household overlap across partitions, while person-permutation deduplication still induces unequal household audit inclusion. Neither the bootstrap nor the Hoeffding illustration repairs that design. We make no US-population claim.

Scientific and computational limits. Only three modest model families, one background realization per dataset, and one state/year of ACS are tested. The MLP fits are not uniformly good predictors: Pima seed 23 has test AUC 0.4826, and German seed 23 has AUC 0.5711; both are retained. Approximation fidelity does not rescue a weak model. No subgroup fairness, temporal transfer, probability calibration, human usefulness, or causal assessment is established. Interventional hybrids may be off-manifold. German's pooled reference leaves unquantified residual target error, and a summed dummy-column TreeSHAP shortcut was rejected because it would change the grouped-player game.

Cached computation makes the controlled audit feasible but is not a latency benchmark. Successful original and expanded campaigns consumed a measured 0.3319 CPU hours in total, including prefix replay; failed-run overhead and historical CPU identity were not recorded. Logical costs are more transferable than these shared timings. We did not evaluate optimized SHAP estimators or Leverage SHAP. The pandas StringDtype failure was corrected and regression-tested before completing German cells; it was an implementation failure, not an unfavorable result removed from analysis (Appendix H).

Implications and next steps. Calling an explanation "reliable" without its metric risks unwarranted confidence in consequential settings. Our findings do not authorize clinical or credit decisions, individual recourse, or a safety interpretation of 0.95 agreement. Priorities are a prespecified equal-probability or design-weighted ACS audit, more independent instances if certification is sought, larger independent German reference pools, and validation of a reference-free budget diagnostic. Each would address a present boundary rather than convert the current descriptive results into a deployment guarantee.

6 Conclusion

A fixed-game audit separates approximation fidelity from the broader validity of an explanation. In the combined 120-instance study, exact-game macro top-three recovery increases from 0.7656 to 0.9656 across the tested budgets, and the held-out next 90 retain this pattern. Yet MLP means remain below the illustrative 0.95 criterion, small reference gaps remain difficult, and strong material-sign agreement does not imply exact leading-feature recovery. German reference agreement and background changes require separate interpretations. The result is evidence for explicit, metric- and game-specific auditing of a single stochastic explanation, not a new estimator, universal certification, or real-world causal or safety validation.

Table 4: Predictive metrics on the entire fixed test partition, not only the audit instances. AUC ↑, log loss ↓, and accuracy ↑ at probability threshold 0.5. These are observed fixed-test metrics; prediction-level intervals cannot be recovered from the aggregate performance table. Empty lists denote no captured fit warning. All fits, including weak MLP fits, are retained. Source: `analysis120/paper_model_performance.tex`.

dataset	model	seed	auc	log_loss	accuracy	fit_warnings
pima	lr	11	0.7606	0.5714	0.6883	[]
pima	lr	23	0.7606	0.5714	0.6883	[]
pima	lr	47	0.7606	0.5714	0.6883	[]
pima	rf	11	0.7509	0.5574	0.6818	[]
pima	rf	23	0.7344	0.5666	0.6818	[]
pima	rf	47	0.7478	0.5618	0.6883	[]
pima	mlp	11	0.7381	0.5622	0.6948	[]
pima	mlp	23	0.4826	0.6483	0.6753	[]
pima	mlp	47	0.7626	0.5668	0.6948	[]
acs	lr	11	0.8364	0.4745	0.7823	[]
acs	lr	23	0.8364	0.4745	0.7823	[]
acs	lr	47	0.8364	0.4745	0.7823	[]
acs	rf	11	0.7863	0.5619	0.6855	[]
acs	rf	23	0.8040	0.5580	0.6815	[]
acs	rf	47	0.8212	0.5517	0.6801	[]
acs	mlp	11	0.8387	0.4861	0.7809	[]
acs	mlp	23	0.8263	0.4801	0.7782	[]
acs	mlp	47	0.8270	0.4818	0.7755	[]
german	lr	11	0.8060	0.4819	0.7700	[]
german	lr	23	0.8060	0.4819	0.7700	[]
german	lr	47	0.8060	0.4819	0.7700	[]
german	rf	11	0.7840	0.5145	0.7550	[]
german	rf	23	0.7963	0.5074	0.7300	[]
german	rf	47	0.8044	0.5086	0.7550	[]
german	mlp	11	0.7604	0.5217	0.7700	[]
german	mlp	23	0.5711	0.6134	0.7100	[]
german	mlp	47	0.7913	0.4955	0.8000	[]

Table 5: Exact-game antithetic macro secondary metrics, generated from `analysis120/exact_macro.csv`. Entries are means [descriptive 95% paired cluster intervals]. Sign recovery and stability are not substitutes for exact top-three recovery.

Metric	$P = 1$	$P = 4$	$P = 16$	$P = 64$
L1 ↓	0.0399 [0.0376, 0.0423]	0.0192 [0.0181, 0.0203]	0.0095 [0.0089, 0.0100]	0.0047 [0.0045, 0.0050]
Normalized L1 ↓	0.1182 [0.1140, 0.1226]	0.0571 [0.0551, 0.0592]	0.0283 [0.0273, 0.0293]	0.0141 [0.0136, 0.0146]
Material sign ↑	0.9983 [0.9980, 0.9986]	0.9999 [0.9998, 0.9999]	1.0000 [1.0000, 1.0000]	1.0000 [1.0000, 1.0000]
Pair match ↑	0.6970 [0.6812, 0.7124]	0.8336 [0.8211, 0.8456]	0.9123 [0.9027, 0.9216]	0.9528 [0.9453, 0.9597]
Stability ↑	0.8395 [0.8308, 0.8477]	0.9151 [0.9087, 0.9213]	0.9557 [0.9508, 0.9603]	0.9762 [0.9725, 0.9797]